

Problem Set 1 – Solutions

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Question 1

Part (i)

For any $z \in \mathbb{C}$ such that $|z| \leq 1$,

$$(1 - z)B(z) = \sum_{\ell=0}^{\infty} B_{\ell}z^{\ell} - \sum_{\ell=0}^{\infty} B_{\ell}z^{\ell+1} = B_0 + \sum_{\ell=1}^{\infty} B_{\ell}z^{\ell} - \sum_{\ell=1}^{\infty} B_{\ell-1}z^{\ell} = B_0 + \sum_{\ell=1}^{\infty} (B_{\ell} - B_{\ell-1})z^{\ell}.$$

By definition, $B_0 = -\sum_{m=1}^{\infty} C_m = C_0 - C(1)$ and $B_{\ell} - B_{\ell-1} = C_{\ell}$ for $\ell \geq 1$. Hence,

$$(1 - z)B(z) = C_0 - C(1) + \sum_{\ell=1}^{\infty} C_{\ell}z^{\ell} = \sum_{\ell=0}^{\infty} C_{\ell}z^{\ell} - C(1) = C(z) - C(1).$$

Part (ii)

By the result in part (i),

$$y_t = C(L)u_t = \{C(1) + (I_n - L)B(L)\}u_t = C(1)u_t + \eta_t - \eta_{t-1},$$

where we define

$$\eta_t \equiv B(L)u_t.$$

This latter series is n -dimensional and covariance stationary because $B(L)$ is absolutely summable and $\{u_t\}$ is covariance stationary.

Question 2

Part (i)

Since $u_t \in \overline{\text{sp}}(y_{\tau}, -\infty < \tau \leq t)$ (by assumption) and $y_t \in \overline{\text{sp}}(u_{\tau}, -\infty < \tau \leq t)$ (by definition of $\{y_t\}$), we have

$$\overline{\text{sp}}(y_{\tau}, -\infty < \tau \leq t) = \overline{\text{sp}}(u_{\tau}, -\infty < \tau \leq t).$$

Thus,

$$\begin{aligned}
\text{Var}^*(y_{t+1} \mid \{y_\tau\}_{-\infty < \tau \leq t}) &= \text{Var}^*(y_{t+1} \mid \{u_\tau\}_{-\infty < \tau \leq t}) \\
&= \text{Var}(y_{t+1} - E^*(y_{t+1} \mid \{u_\tau\}_{-\infty < \tau \leq t})) \\
&= \text{Var}\left(\sum_{\ell=0}^{\infty} \Theta_\ell u_{t+1-\ell} - \sum_{\ell=1}^{\infty} \Theta_\ell u_{t+1-\ell}\right) \\
&= \text{Var}(\Theta_0 u_{t+1}) \\
&= \text{Var}(u_{t+1}) \\
&= \sigma_u^2.
\end{aligned}$$

Part (ii)

For the same reason as in part (i), we have

$$\overline{\text{sp}}(\{y_\tau, x_\tau\}, -\infty < \tau \leq t) = \overline{\text{sp}}(\{u_\tau, x_\tau\}, -\infty < \tau \leq t).$$

Thus,

$$\begin{aligned}
\text{Var}^*(y_{t+1} \mid \{y_\tau, x_\tau\}_{-\infty < \tau \leq t}) &= \text{Var}^*(y_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t}) \\
&= \text{Var}(y_{t+1} - E^*(y_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t})) \\
&= \text{Var}\left(\sum_{\ell=0}^{\infty} \Theta_\ell u_{t+1-\ell} - E^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t}) - \sum_{\ell=1}^{\infty} \Theta_\ell u_{t+1-\ell}\right) \\
&= \text{Var}(u_{t+1} - E^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t})) \\
&= \text{Var}^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t}).
\end{aligned}$$

Part (iii)

Define $\tilde{u}_{t+1} \equiv u_{t+1} - E^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t})$. By definition of the best linear predictor, $\text{Cov}(\tilde{u}_{t+1}, E^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t})) = 0$. Thus,

$$\begin{aligned}
\sigma_u^2 &= \text{Var}(u_{t+1}) \\
&= \text{Var}(\tilde{u}_{t+1} + E^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t})) \\
&= \text{Var}(\tilde{u}_{t+1}) + \text{Var}(E^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t}))
\end{aligned}$$

$$= \text{Var}^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t}) + \text{Var}(E^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t})),$$

which is known as the “law of total variance”. Clearly, we have

$$\sigma_u^2 = \text{Var}^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t})$$

if and only if

$$\text{Var}(E^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t})) = 0,$$

i.e., if and only if there exists $c \in \mathbb{R}$ such that

$$E^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t}) = c \quad \text{almost surely.}$$

Since all variables have mean zero, we must in fact have $c = 0$, if it exists. But

$$E^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t}) = 0$$

if and only if

$$\text{Cov}(u_{t+1}, u_{t-\ell}) = \text{Cov}(u_{t+1}, x_{t-\ell}) = 0 \quad \text{for all } \ell \geq 0,$$

by the orthogonality conditions for best linear predictors. Since u_t is white noise, we automatically have $\text{Cov}(u_{t+1}, u_{t-\ell}) = 0$ for all $\ell \geq 0$. This proves the statement in the problem set.

Part (iv)

Note first that $E^*(x_t \mid \{y_\tau\}_{-\infty < \tau \leq t+\ell}) = E^*(x_t \mid \{u_\tau\}_{-\infty < \tau \leq t+\ell})$, as argued in part (i). We now check the orthogonality conditions for the latter best linear predictor. For any $\tau \leq t+\ell$,

$$\begin{aligned} \text{Cov}\left(x_t - \sum_{m=-\infty}^{\ell} a_m u_{t+m}, u_\tau\right) &= \text{Cov}(x_t, u_\tau) - \sum_{m=-\infty}^{\ell} a_m \text{Cov}(u_{t+m}, u_\tau) \\ &= \text{Cov}(x_t, u_\tau) - a_{\tau-t} \text{Cov}(u_\tau, u_\tau) \\ &= \text{Cov}(x_t, u_\tau) - \frac{\text{Cov}(x_t, u_\tau)}{\sigma_u^2} \sigma_u^2 \\ &= 0, \end{aligned}$$

where the penultimate equality uses the definition of a_m . Based on the above orthogonality result, we conclude that the statement in the problem set is true.

Part (v)

By the law of total variance,

$$\text{Var}^*(x_t \mid \{y_\tau\}_{-\infty < \tau < \infty}) = \text{Var}^*(x_t \mid \{y_\tau\}_{-\infty < \tau \leq t})$$

if and only if

$$\text{Var}(E^*(x_t \mid \{y_\tau\}_{-\infty < \tau < \infty})) = \text{Var}(E^*(x_t \mid \{y_\tau\}_{-\infty < \tau \leq t})),$$

which by part (iv) holds if and only if

$$\sum_{m=-\infty}^{\infty} \text{Cov}(x_t, u_{t+m})^2 = \sum_{m=-\infty}^0 \text{Cov}(x_t, u_{t+m})^2,$$

which holds if and only if

$$\text{Cov}(x_t, u_{t+m}) = 0 \quad \text{for all } m \geq 1,$$

which by part (iii) holds if and only if

$$\sigma_u^2 = \text{Var}^*(u_{t+1} \mid \{u_\tau, x_\tau\}_{-\infty < \tau \leq t}),$$

which by parts (i) and (ii) is equivalent with

$$\text{Var}^*(y_{t+1} \mid \{y_\tau\}_{-\infty < \tau \leq t}) = \text{Var}^*(y_{t+1} \mid \{y_\tau, x_\tau\}_{-\infty < \tau \leq t}).$$

Questions 3 and 4

See the Matlab script `ps1_solution.m` on the course website, along with the associated data set and functions. (Note: Unlike me, you are expected to explain your code, report relevant numbers/figures, and discuss the results in your assignment.)